Compactness through finite intersections

Theorem 2.4: Compactness by closed sets

All levels · read along the `refines` order

Snapshot 1671ccb1e155

Contents

2.4 Compactness by closed sets **1**

2.5 Proof of Theorem 2.4 1

Outside this export **1**

Theorem 2.4 (Compactness by closed sets)**.** *X is compact if and only if every family $(F_i)_{i\in I}$ of closed subsets of X with the finite intersection property has $\bigcap_{i\in I} F_i \neq \emptyset$.*

Proof 2.5. Of Compactness by closed sets (Theorem 2.4).

Both directions are the contrapositive of each other once the complements are taken: Lemma 2.6 exchanges the two conditions, and Lemma 2.7 says what a finite subcover becomes.

USES → Finite intersection property (Definition 2.2*)
Where the finite intersection property enters. Only in Lemma 2.7: the property is the statement that no finite subfamily has empty intersection, word for word.

Lemma 2.6 (Complements)**.** *For a family $(F_i)_{i\in I}$ of closed sets put $U_i = X \setminus F_i$. Then*

$$\bigcap_{i\in I} F_i = \emptyset \iff \bigcup_{i\in I} U_i = X,$$

so (U_i) is an open cover exactly when the F_i have no common point.

Lemma 2.7 (Finite subfamilies)**.** *For $J \subseteq I$ finite, $(U_j)_{j\in J}$ covers X if and only if $\bigcap_{j\in J} F_j = \emptyset$. Hence (U_i) has no finite subcover if and only if (F_i) has the finite intersection property.*

■

Outside this export

References marked * are to parts of the blueprint this document leaves out:

Definition 2.2 Finite intersection property, in 2 Covers and compact spaces (`def-fip`)